Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Subject: Selection Functionals Paper - My Response to Your Latest Comments

Benjamin D. Summers <bsummers@adagiollc.com> Mon, Sep 28, 2026 at 8:38 PM
To: Ezequiel Boero <ezequiel.boero@adagioinstitute.org>
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Thank you, Ezequiel. Revision 3.1 is on SSRN (https://dx.doi.org/10.2139/ssrn.6922758) with the three amendments and nothing else; every section and result number is unchanged.

Item 1. I took the exact condition. Proposition 7.5 now reads "let the proposal and surrogate be as in Corollary 3.6 with $k_\Pi/\sigma_\Pi^2 + \lambda k_\nu/\sigma_\nu^2 \geq 0$, as when both rates are positive". The proof states that $k_\bullet \delta t < 1$ gives $a_\Pi, a_\nu > 0$ and hence $u > 0$ for either sign of the rates, that $E \succeq 0$ follows from $\kappa \geq 0$ and $\kappa_2 \geq 0$, and that the hypothesis is necessary, with your witness: the last coordinate vector e has $Le = 0$, so $e^{\intercal}Ee = \delta t\kappa$. Your case (a) reproduces ($a_\Pi = 1.030$, $a_\nu = 1.020$, $u = 1.3318$, $\kappa = -3.5917$), and there E has least eigenvalue -1.28 and the bound $C_M \preceq (\delta t/u)I$ fails together with both trace bounds. Your three pairs in (c) satisfy every bound at $M = 100$, 400 and 1600. Inside the paper's surrogate family, whose rate is positive by definition, the hypothesis admits a mildly explosive proposal: $k_\Pi = -0.5$ and $k_\Pi = -1$ with $k_\nu = 2$ at $\lambda = 1/2$ ($\kappa = 0.5$ and 0) satisfy every bound, and $k_\Pi = -0.9$ with $k_\nu = 1$ ($\kappa = -0.4$) fails the operator bound. Corollary 3.6 is unchanged; none of its claims uses the sign.

Item 2. Table 3 carries $+ O(\delta t)$.

Item 3. Section 10.1 now says that in the Gaussian control of Section 9.4 the ratio has bounded chi-square divergence under refinement, and that in general a likelihood ratio lies in L^1 of the reference measure and not necessarily in L^2, so boundedness requires a uniform second-moment condition on the densities; for the stochastic exponential the second moment is $E_{Q'}[\exp\int\theta^2 dt]$ under the drift-2θ measure, finite under a Novikov-type condition on $\int\theta^2 dt$. The χ^2 row of Table 3 carries the same qualification. Your counterexample checks: $x^{-1/2}/2$ has mass one and second moment $(1/4)\ln(1/\varepsilon)$ on $[\varepsilon, 1]$.

Thank you for the check on the recursion.

Ben

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